PRACTICAL PROJECT REPORT

Automotive CRM Lead Conversion Intelligence, Predictive ML & AI Copilot

Entity:	Bhubaneswar Kia Dealership	**Author / Lead:**	Anmol Kumar (Anmolkr001)
Dataset Scope:	6,967 Validated Leads (Jan–Jun 2026)	**Live Vercel URL:**	kia-data-analytics.vercel.app
Core Stack:	React 19, FastAPI, XGBoost, LangChain	**GitHub Repo:**	github.com/Anmolkr001/KIA-Data-Analytics

1. Executive Summary

This project engineers an end-to-end commercial intelligence platform for the **Bhubaneswar Kia Dealership**. Modern automotive retail suffers from high digital acquisition spend coupled with poor downstream lead visibility. By ingesting and analyzing **6,967 actual CRM leads** across 86 attributes, this system introduces: (1) rigorous empirical funnel attribution, (2) an XGBoost classification engine with local SHAP feature attributions, (3) a sandboxed natural-language query agent (LangChain AI Copilot), and (4) an interactive React 19 web dashboard with an immersive 360° vehicle showroom, strictly secured by credential gates.

Key Highlights:
- **Overall Conversion: 6,967 Enquiries resulted in 786 Retail Deliveries (11.28% Conversion Rate).**
- **Test Drive Multiplier: Prospects completing a Test Drive convert at 19.11% vs. 4.44% without (4.30× lift).**
- **Follow-up Sweet Spot: Optimal retail conversion peaks at 2 to 4 follow-ups (19.06%); collapses to <1.6% past 5 calls.**
- **ML Predictor Accuracy: XGBoost achieves ROC-AUC of 0.9804 with 97.8% recall on retail conversion prospects.**

2. Technical Architecture & Data Pipeline

The application is organized into four decoupled architectural tiers ensuring speed, sandbox safety, and clean separation of concerns:

- **Data Engineering Layer:** Raw CRM exports (6,967 rows × 86 columns) are cleaned via src/clean_data.py, normalized for phone formatting, postal codes, and sales cycle timing, and mirrored into a local SQLite database (data/leads.db).
- **Machine Learning Engine:** An XGBClassifier trained on 15 demographic, behavioral, and engagement features, addressing the 7.86:1 negative-to-positive class imbalance via cost-sensitive learning.
- **Microservice Backend:** FastAPI running on port 8000 exposes REST endpoints for analytical trends, paginated lead tables, real-time ML inference (POST /predict), and natural language LangChain execution.
- **Client Experience Layer:** React 19 single-page application built on Vite and TailwindCSS, deployed globally on Vercel with role-authenticated access, 8 analytical views, and 3D car showroom.

3. Empirical CRM Analytical Findings

Quantitative analysis across all stages of the customer acquisition lifecycle yielded distinct empirical benchmarks:

3.1 Full Funnel Conversion & Drop-off Rates

Funnel Stage	Total Leads	Stage Conversion	Cumulative Conv.	Drop-off Rate
Enquiry	6,967	100.0%	**100.0%**	0.0%
Test Drive	3,249	46.63%	**46.63%**	53.37%
Booking	721	22.19%	**10.35%**	77.81% (Peak Drop)
Retail Delivery	786	109.02%*	**11.28%**	Fulfillment >95%

*Retail deliveries exceed active bookings due to direct walk-in spot deliveries and pre-booked backlog rollovers.

3.2 Test Drive Impact Analysis

Test Drive Completed	Enquiries	Bookings	Retail Units	Retail Conversion Rate
Yes (Completed)	3,249	579	621	**19.11% (4.30x Multiplier)**
No (Not Done)	3,718	142	165	**4.44%**

3.3 Lead Sourcing Channel Efficiency vs. Volume

Channel	Enquiries	Share %	Test Drives	Retails	TD Rate	Retail Conv. Rate
Referral	1,097	15.7%	645	303	58.80%	**27.62% (Highest)**
Walk-in	1,627	23.4%	1,191	219	73.20%	**13.46%**
Field Marketing	941	13.5%	610	122	64.82%	**12.96%**
Telephone Direct	384	5.5%	191	29	49.74%	**7.55%**
Online / Central CRM	676	9.7%	197	36	29.14%	**5.33%**
Hyperlocal Paid Ads	2,232	32.0%	412	77	18.46%	**3.45% (Lowest)**

3.4 Vehicle Model Commercial Demand & Conversion

Vehicle Model	Enquiries	Test Drives	Bookings	Retail Deliveries	Retail Conversion
SONET	2,115	1,123	279	344	**16.26% (Top Anchor)**
CARENS	1,326	656	159	195	**14.71%**
NEW SELTOS	2,155	1,241	226	187	**8.68%**
SYROS	332	87	18	16	**4.82%**
CARENS CLAVIS EV	441	110	17	20	**4.54%**
SELTOS (Legacy)	566	32	22	24	**4.24%**
CARNIVAL / EV6 / EV9	32	0	0	0	**0.00%**

3.5 Consultant Follow-Up Cadence & Diminishing Returns

Follow-up Bucket	Enquiries	Retail Deliveries	Conversion Rate	Operational Status
0 Calls	372	0	**0.00%**	Uncontacted Leads (Leakage)
1 Call	1,591	5	**0.31%**	Under-contacted
2 Calls	2,203	420	**19.06%**	Peak Conversion Window
3 – 4 Calls	2,034	351	**17.26%**	High Efficiency Window
5 – 6 Calls	487	8	**1.64%**	Diminishing Returns (Drop 90%)
7+ Calls	280	2	**0.71%**	Unviable / Reassign to Drip

4. Predictive Machine Learning & Explainable AI

To prioritize incoming leads dynamically, an **Extreme Gradient Boosting (XGBoost)** model was trained to predict the probability of retail delivery. Stratified 5-Fold Cross Validation was employed to prevent data leakage.

Metric	Cross-Validation (Mean ± Std)	Holdout Test Set	Operational Interpretation
ROC-AUC	0.9771 ± 0.0039	**0.9804**	Near-perfect ranking discrimination
PR-AUC (Avg Precision)	0.8377	**0.8642**	High precision in skewed positive class
Recall (Lead Capture)	0.9784	**0.9810**	Captures 98.1% of all true buyers
F1 Score	0.7057	**0.7214**	Harmonic balance of precision and recall

Top SHAP Local Feature Attribution Factors:

1. lead_warmth_Hot (+3.08 SHAP): Primary initial positive signal for confirmed intent.
2. followup_count (+0.49 SHAP): Active two-way engagement strongly drives conversion probability.
3. exchange_interest_Yes (+0.40 SHAP): Trade-in intent provides an immediate 4.5× conversion lift.
4. customer_type_Corporate (+0.36 SHAP): B2B fleet approvals convert at 80.86%.
5. aging_days (+0.27 SHAP): Indicates sustained showroom negotiation and financing approvals.

5. LangChain AI Copilot & Security Sandbox

The conversational assistant enables dealership personnel to query the database using plain natural language. To ensure production stability, an **AST Static Security Validator** strictly disallows code injection, filesystem modification, and infinite loops, enforced with a 5-second execution deadline.

Benchmark Category	Questions Tested	Accuracy (%)	Avg. Latency (ms)	Test Result
Funnel KPIs	5	**100.0%**	262.4 ms	All 5 Passed
Sourcing Channels	3	**100.0%**	145.2 ms	All 3 Passed
Funnel Impact (Test Drives)	2	**100.0%**	178.6 ms	All 2 Passed
Vehicle Models	2	**100.0%**	148.9 ms	All 2 Passed
Geography (Bhubaneswar)	1	**100.0%**	143.2 ms	Passed
CRM Operations (Follow-ups)	2	**100.0%**	157.6 ms	All 2 Passed
Overall Test Suite	15	**100.0%**	190.8 ms	15 / 15 Passed (Zero Errors)

6. Actionable Managerial & Strategic Directives

1. Enforce Mandatory Test Drive Scheduling within 4 Hours: With test-drives yielding a 4.3× conversion multiplier, dealership management should set a KPI targeting 60% test-drive completion. This alone would generate an estimated +180 incremental retail sales annually.

2. Restructure Digital Sourcing Budgets: Reduce low-intent Hyperlocal ad spend (3.45% conversion) by 25% and reallocate those funds into customer referral reward programs (27.62% conversion) and dedicated B2B fleet outreach (80.86% conversion).

3. Automate Cadence Rules after 4 Follow-ups: Reassign leads with 5+ unsuccessful follow-ups to automated WhatsApp/email nurturing sequences, preventing consultant burnout and redirecting attention to high-scoring prospects.

4. Launch an Instant Trade-in Appraisal Counter: Offer free 10-minute certified vehicle valuations on every showroom entry; capturing trade-in intent boosts conversion from 6.11% to 27.61%.

Document Information: Generated for technical and operational evaluation. Interactive dashboard live at **kia-data-analytics.vercel.app**. Full source repository: **github.com/Anmolkr001/KIA-Data-Analytics**.